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toy_imputation.R

```

1 my_mat<-data.frame(SNP1=c(0,1,1,0,1),SNP2=c(0,2,0,1,0),SNP3=c(0,1,0,0,1))
2 rownames(my_mat)<-c("Ind1","Ind2","Ind3","Ind4","Ind5")
3 my_mat
4
5 my_mat_with_miss<-my_mat
6 my_mat_with_miss[3,3]<-NA
7 my_mat_with_miss
8
9 summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1))$adj.r.squared
10 summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP2))$adj.r.squared
11
12 as.integer(predict(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1), newdata = data.frame(SNP3=my_mat_with_miss$SNP3[3]) ))
13

```

13:1 (Top Level)

R Script

Console Terminal Background Jobs

R 4.3.3 ~/

```

>
>
> my_mat<-data.frame(SNP1=c(0,1,1,0,1),SNP2=c(0,2,0,1,0),SNP3=c(0,1,0,0,1))
> rownames(my_mat)<-c("Ind1","Ind2","Ind3","Ind4","Ind5")
> my_mat
  SNP1 SNP2 SNP3
Ind1  0    0    0
Ind2  1    2    1
Ind3  1    0    0
Ind4  0    1    0
Ind5  1    0    1
> my_mat_with_miss<-my_mat
> my_mat_with_miss[3,3]<-NA
> my_mat_with_miss
  SNP1 SNP2 SNP3
Ind1  0    0    0
Ind2  1    2    1
Ind3  1    0  NA
Ind4  0    1    0
Ind5  1    0    1
> summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1))$adj.r.squared
[1] 1

```

Warning message:

```
In summary.lm(lm(my_mat_with_miss$SNP3 ~ my_mat_with_miss$SNP1)) :
essentially perfect fit: summary may be unreliable
```

```

> summary(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP2))$adj.r.squared
[1] -0.3636364
> as.integer(predict(lm(my_mat_with_miss$SNP3~my_mat_with_miss$SNP1), newdata = data.frame(SNP3=my_mat_with_miss$SNP3[3]) ))
[1] 0 1 1 0 1

```

Warning message:

```
'newdata' had 1 row but variables found have 5 rows
```

>

Posit Assistant Project: (None)

Environment History Connections Tutorial

Import Dataset 293 MiB

R Global Environment

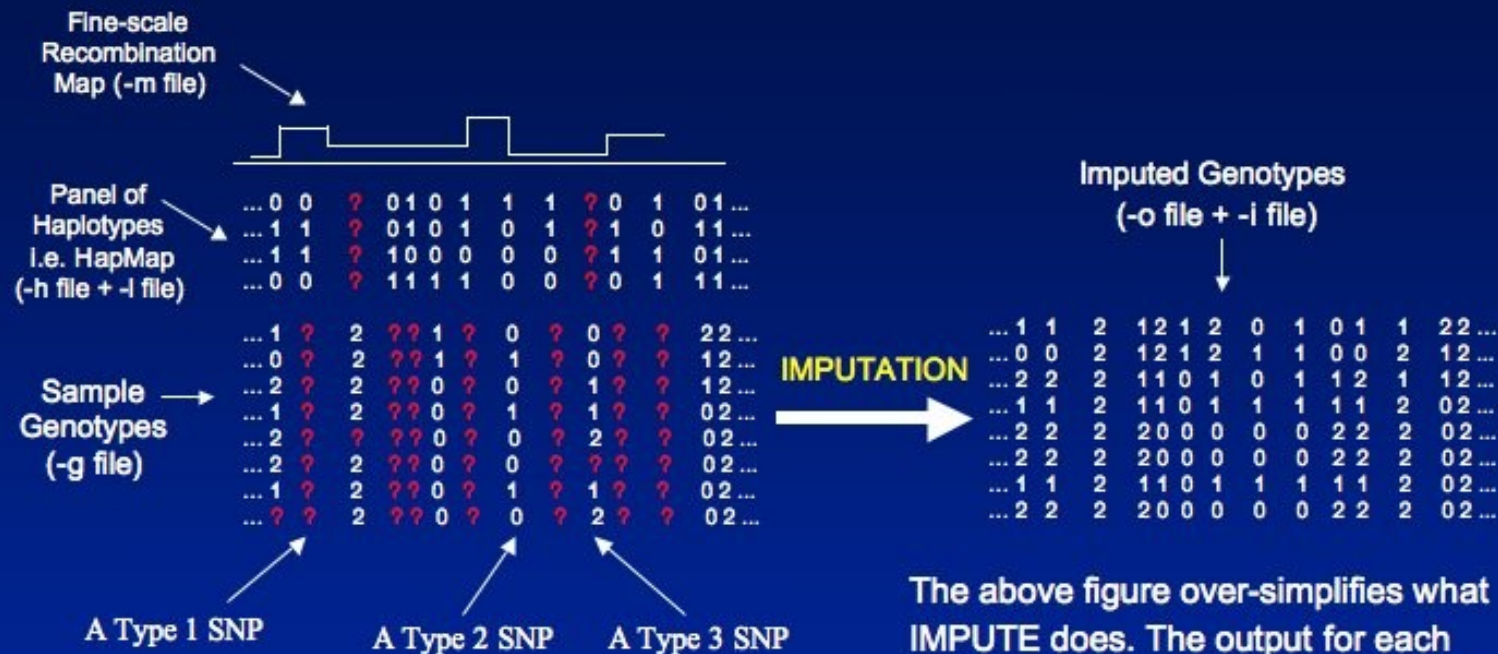
Data

my_mat	5 obs. of 3 variables
my_mat_with_miss	5 obs. of 3 variables

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Zoom Export

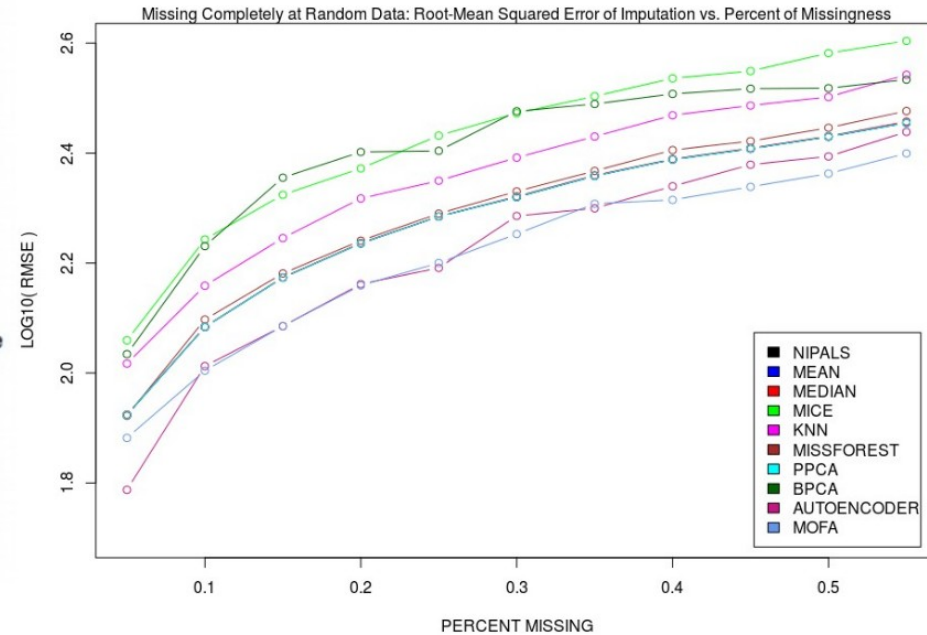
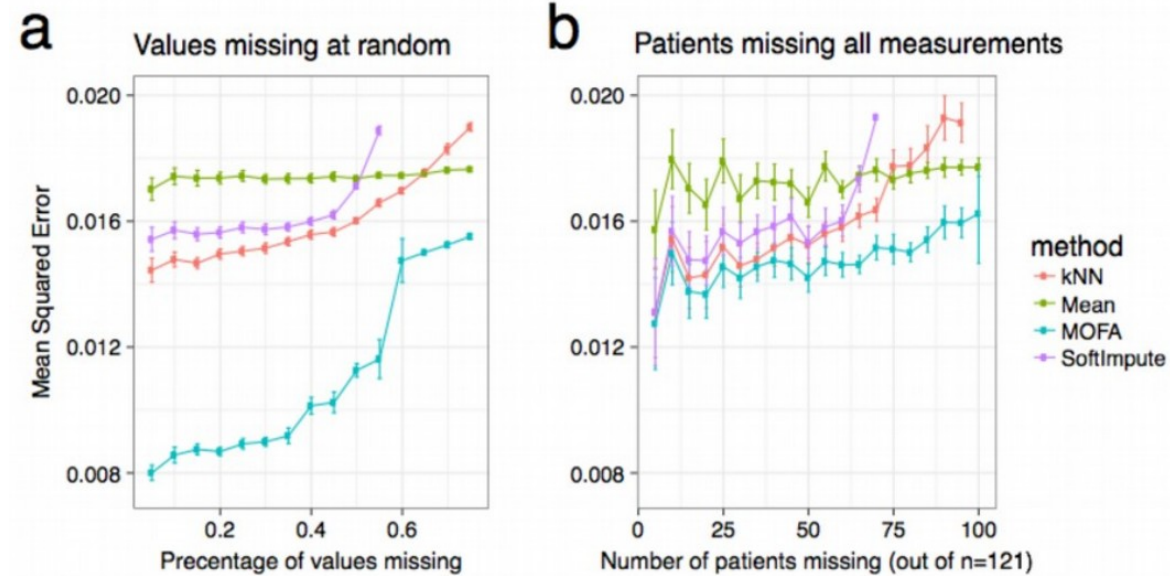
Overview of Imputation using IMPUTE



The above figure over-simplifies what IMPUTE does. The output for each genotype is a probability distribution on genotypes i.e.

0	1	2
0.01	0.18	0.81

This captures the uncertainty in the prediction.



Bayesian framework is insensitive to missing data, priors compensate for the lack of data

IMPUTE2

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Last update: 23 Dec 2014

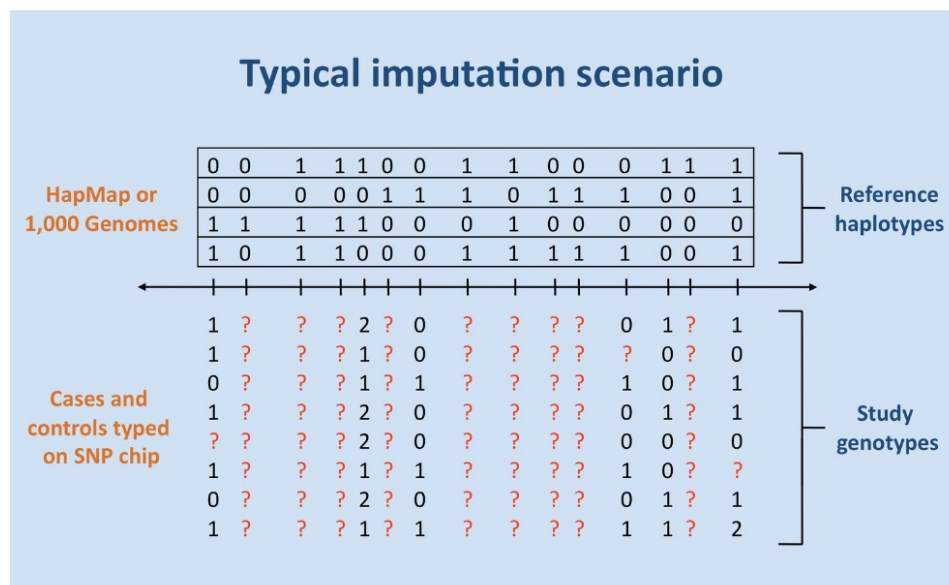
IMPUTE2

IMPUTE version 2 (also known as **IMPUTE2**) is a genotype imputation and haplotype phasing program based on ideas from Howie *et al.* 2009:

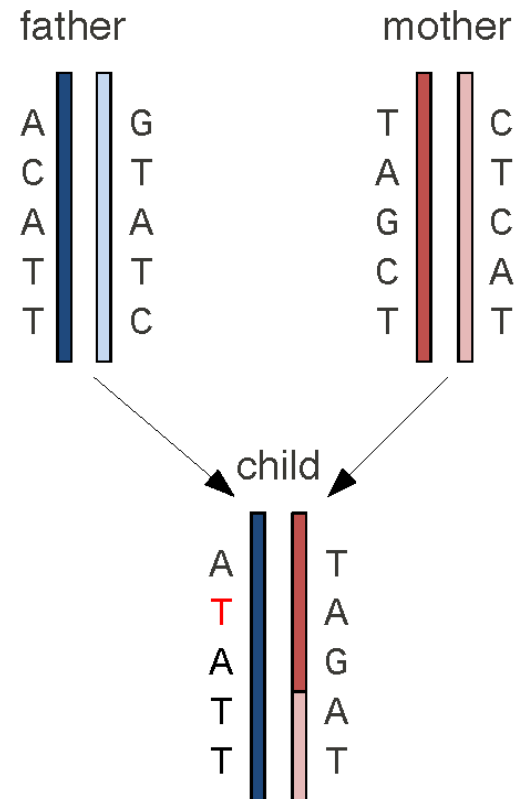
B. N. Howie, P. Donnelly, and J. Marchini (2009) **A flexible and accurate genotype imputation method for the next generation of genome-wide association studies**. *PLoS Genetics* 5(6): e1000529 [[Open Access Article](#)] [[Supplementary Material](#)]

IMPUTE2 also includes features that were introduced in other publications, which you can find [here](#).

The figure below shows the most common scenario in which imputation is used: unobserved genotypes (red question marks) in a set of study individuals are imputed (or predicted) using a set of reference haplotypes and genotypes from a SNP chip.



Phasing



SHAPEIT

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Last update: 17 January 2018

SHAPEIT

le cnam



SHAPEIT is a fast and accurate method for [estimation of haplotypes](#) (aka phasing) from genotype or sequencing data.

SHAPEIT has primarily been developed by [Dr Olivier Delaneau](#) through a collaborative project between the research groups of [Prof Jean-Francois Zagury](#) at [CNAM](#) and [Prof Jonathan Marchini](#) at [Oxford](#). Funding for this project has been received from several sources : [CNAM](#), [Peptinov](#), [MRC](#), [Leverhulme](#), [The Wellcome Trust](#).

SHAPEIT has several notable features:

- Linear complexity with the number of SNPs and conditioning haplotypes.
- Whole chromosome [GWAS](#) scale datasets can be phased in a single run.
- Phasing individuals with any level of relatedness
- Phasing is multi-threaded to tailor computational times to your resources.
- Handles X chromosomes phasing.
- Phasing using a reference panel (eg.1,000 Genomes) to aid phasing
- Ideal for pre-phasing imputation together with [IMPUTE2](#)

SHAPEIT is free for academic use only. For commercial use please see [here](#)

Beagle 5.5

Program: beagle 5.5 (version: 27Feb25.75f)

Author: Brian Browning

Email: browning@uw.edu

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Introduction

Beagle is a software package for phasing genotypes and imputing ungenotyped markers. Beagle version 5.5 has improved memory efficiency and accuracy when analyzing large sequence data sets.

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Citation

If you use Beagle in a published analysis, please report the program version and cite the appropriate article.

The Beagle 5.5 genotype phasing method is described in:

B L Browning, X Tian, Y Zhou, and S R Browning (2021) Fast two-stage phasing of large-scale sequence data. *Am J Hum Genet* 108(10):1880-1890. doi:10.1016/j.ajhg.2021.08.005

The Beagle 5.5 genotype imputation method is described in:

B L Browning, Y Zhou, and S R Browning (2018). A one-penny imputed genome from next generation reference panels. *Am J Hum Genet* 103(3):338-348. doi:10.1016/j.ajhg.2018.07.015

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Download Beagle 5.5

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ARTICLE

A One-Penny Imputed Genome from Next-Generation Reference Panels

Brian L. Browning,^{1,2,*} Ying Zhou,² and Sharon R. Browning²

Genotype imputation is commonly performed in genome-wide association studies because it greatly increases the number of markers that can be tested for association with a trait. In general, one should perform genotype imputation using the largest reference panel that is available because the number of accurately imputed variants increases with reference panel size. However, one impediment to using larger reference panels is the increased computational cost of imputation. We present a new genotype imputation method, Beagle 5.0, which greatly reduces the computational cost of imputation from large reference panels. We compare Beagle 5.0 with Beagle 4.1, Impute4, Minimac3, and Minimac4 using 1000 Genomes Project data, Haplotype Reference Consortium data, and simulated data for 10k, 100k, 1M, and 10M reference samples. All methods produce nearly identical accuracy, but Beagle 5.0 has the lowest computation time and the best scaling of computation time with increasing reference panel size. For 10k, 100k, 1M, and 10M reference samples and 1,000 phased target samples, Beagle 5.0's computation time is 3× (10k), 12× (100k), 43× (1M), and 533× (10M) faster than the fastest alternative method. Cost data from the Amazon Elastic Compute Cloud show that Beagle 5.0 can perform genome-wide imputation from 10M reference samples into 1,000 phased target samples at a cost of less than one US cent per sample.

Introduction

Genotype imputation uses a reference panel of phased, sequenced individuals to estimate sequence data in target samples that have been genotyped on a SNP array.^{1,2} Imputation of ungenotyped markers is a standard tool in genome-wide association studies because it greatly increases the number of markers that can be tested for association with a trait. Genotype imputation also provides a foundation for meta-analysis of genome-wide association studies because it converts data for samples that have been genotyped on different SNP arrays into genotype data for a shared set of sequence variants. Meta-analyses based on genotype imputation have discovered thousands of new genetic associations.³

The reference panel plays the primary role in determining the accuracy of imputed variants. Imputation accuracy for a variant generally increases with increasing reference panel size, and variants must be present in the reference panel in order to be accurately imputed.^{4,5} Consequently, whenever a significantly larger reference panel becomes available, it is advantageous to re-impute target samples with the larger panel for subsequent analysis.

The value of larger reference panels has led to steadily increasing reference panel size, with the largest reference panels to date having tens of thousands of samples.⁶⁻¹² Ongoing, large-scale projects, such as the Trans-Omics for Precision Medicine (TopMed) program¹³ and the National Human Genome Research Institute's Centers for Common

Increasing reference panel size increases the computational cost of imputation, and this increasing cost has motivated the development of many new computational methods and optimizations. Current imputation methods are able to make use of a rich palette of computational techniques, including the use of identity-by-descent,^{6,14,15} haplotype clustering,^{4,16} and linear interpolation⁵ to reduce the model state space, the use of pre-phasing to reduce computational complexity,^{14,17,18} and the use of specialized reference file formats to reduce file size and memory footprint.^{4,5,19} The techniques developed thus far have made it possible to provide imputation using reference panels with tens of thousands of individuals as a free web service.⁴

However, the total computational cost of imputation is substantial and increasing. Millions of individuals have been genotyped on SNP arrays and it is probable that millions more will be genotyped in the future. A sample may be re-imputed to a current state-of-the-art reference panel each time it is analyzed or included in a meta-analysis. The increasing size of reference panels increases the per-sample cost of re-imputation. If the past exponential growth in reference panel size continues, reference panels could have ten million individuals within a decade.^{6,7,10-12}

In this paper, we present new methodology that reduces the computational cost of genotype imputation 500-fold when imputing from 10M reference samples, and we use the Amazon Elastic Compute Cloud to show that the new methodology can impute genotypes from 10M reference samples into 1,000 phased target samples for a cost of less than one US penny per sample.

BEAGLE reference dataset at genomic positions that were not found in the original 1000 Genomes dataset. Upon further investigation, most of these could be attributed to one-off position differences with some indels reported in the 1000 Genomes dataset. This demonstrates how different data processing pipelines handle complex genotype information in slightly different ways. For the same reason, the total number of markers at unique positions differs in each version of the reference dataset.

Results

All programs outperformed others in certain areas. Based on all of the metrics measured, IMPUTE2 seemed to perform with the greatest accuracy and quality although other programs performed better in other areas.

As expected, pre-phasing the original dataset drastically improved the total compute time.

When the data was pre-phased, IMPUTE2 ran the quickest, followed by Minimac, and then BEAGLE. Without pre-phasing, IMPUTE2 was much faster than BEAGLE.

IMPUTE2 also had superior concordance rates, although all software programs performed well in this area. Minimac had the lowest concordance rate at 96.25%.

Software	Total Compute Time*	Mean SNP Concordance	Total # SNPs	# High Quality SNPs	% High Quality Imputed
IMPUTE2	23 hours	99.98%	668,180	620,792	92.9%
BEAGLE	213 hours	98.43%	484,023	320,991	66.3%
IMPUTE2 with Pre-phasing	8 hours	99.92%	668,180	297,196	44.5%
BEAGLE with Pre-phasing	34 hours	98.05%	484,023	293,890	60.7%
Minimac	18 hours	96.25%	667,870	450,790	67.5%

**includes all steps required*

Without pre-phasing, IMPUTE2 had the highest quality imputation, but after pre-phasing, the certainty metric provided in the IMPUTE2 output dropped dramatically (see first figure below). The R^2 accuracy value given by BEAGLE was also lower in the output based on pre-phased data, but the change was not nearly as dramatic (see